Centro

de

Procesamiento

de

Datos

Práctica 7. Copias de seguridad

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Introducción

Vagrantfile

La máquina virtual que voy a usar es un ubuntu 18.04 con el siguiente vagrantfile:

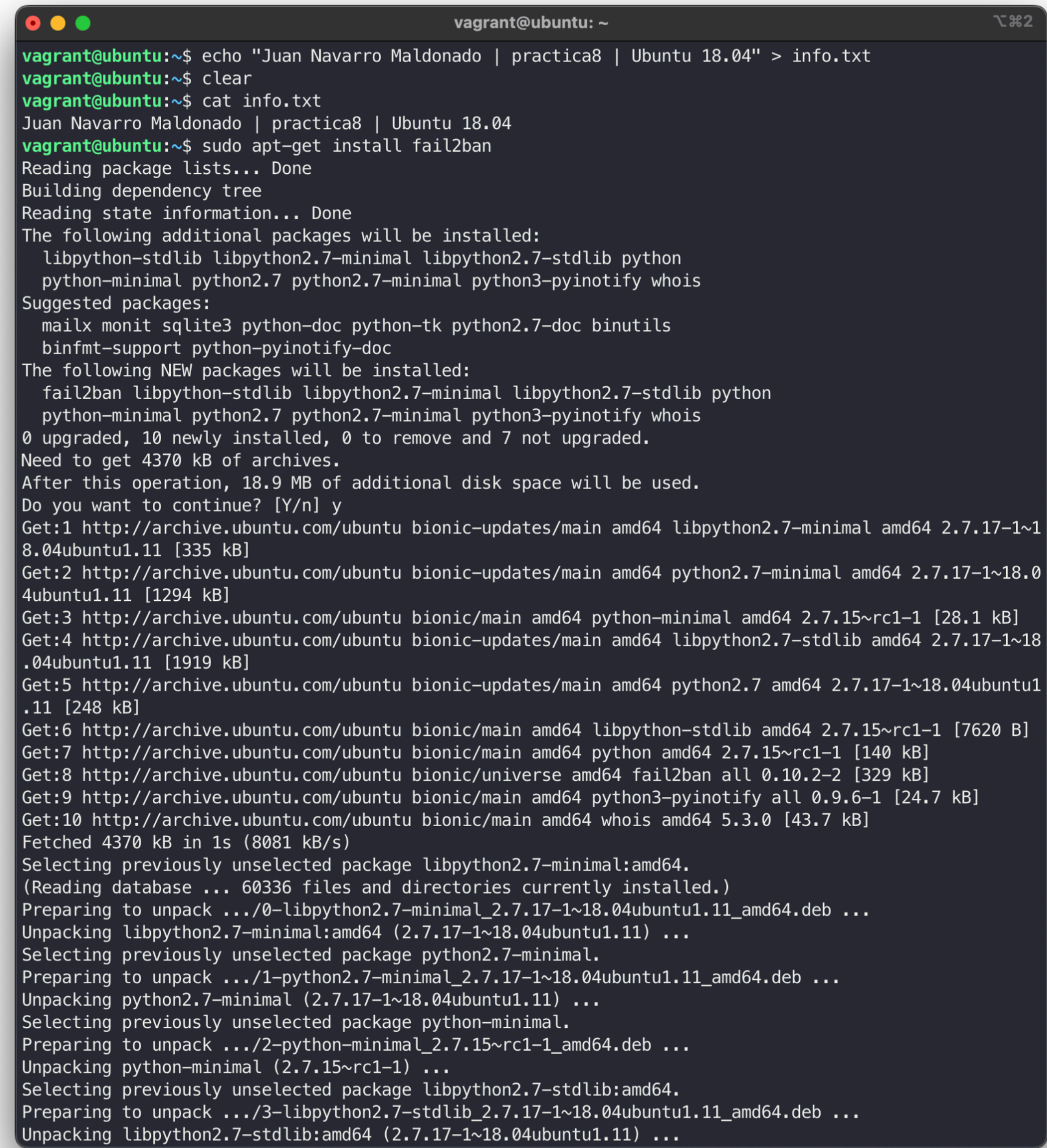
```
Vagrant.configure("2") do |config|
  config.vm.box = "ubuntu/bionic64"  # Ubuntu 18.04
  config.vm.network "private_network", type: "dhcp"

  config.vm.hostname = "ubuntu"
  config.vm.provision "shell", inline: <<-SHELL
    sudo apt-get update -y
    sudo apt-get upgrade -y
  SHELL

  config.vm.network "forwarded_port", guest: 80, host: 8080
  config.vm.network "forwarded_port", guest: 443, host: 8443
  config.vm.provider "virtualbox" do |vb|
    vb.memory = "2048"
    vb.cpus = 4
  end
end
```

1.- Fail2ban

Instalación de fail2ban en Ubuntu

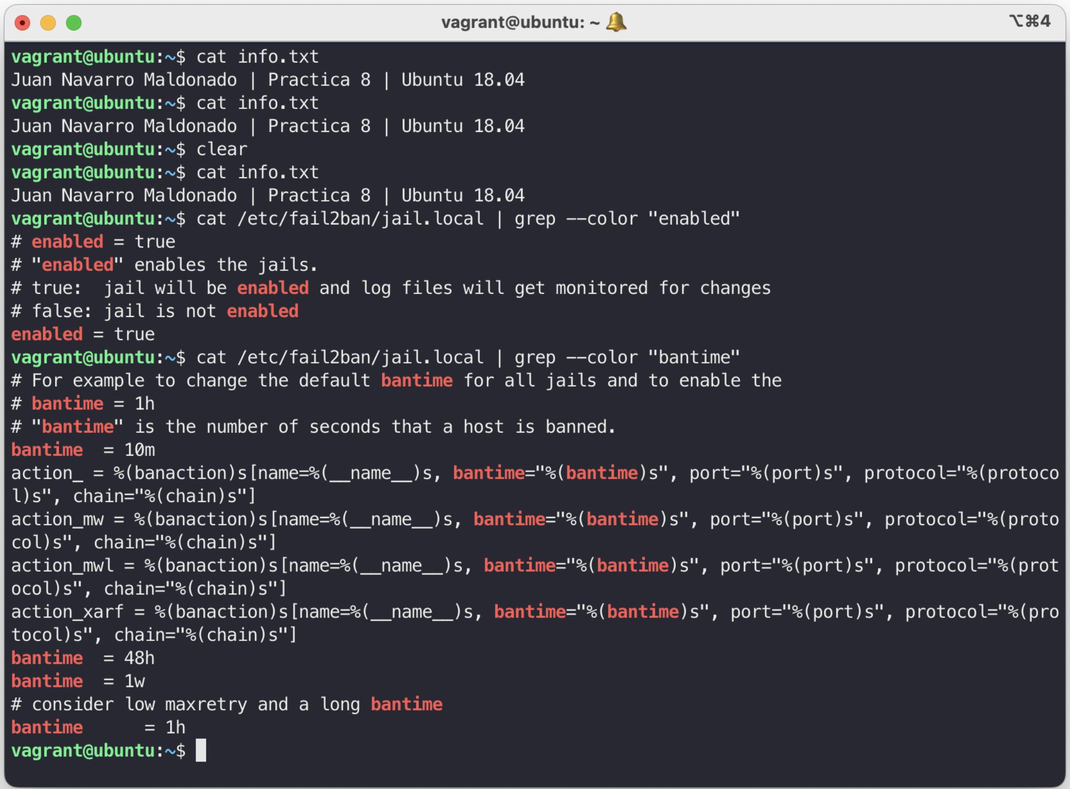
Instalaremos Fail2Ban con el siguiente comando:

`sudo apt-get install fail2ban`



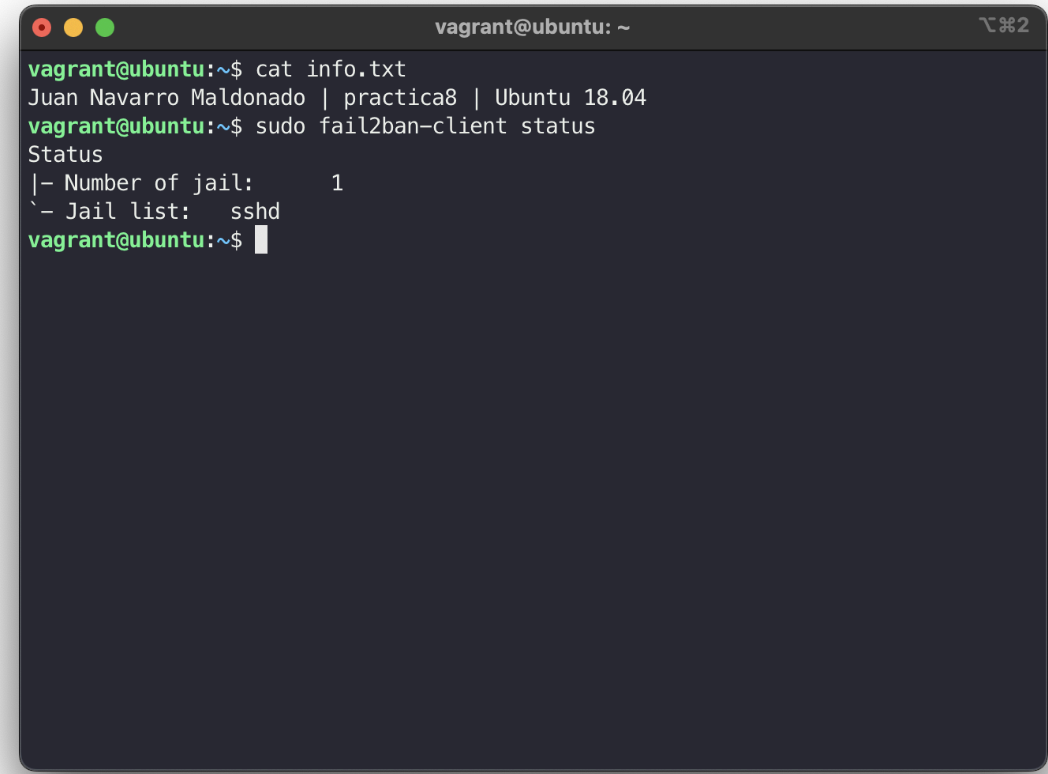
```
vagrant@ubuntu: ~
vagrant@ubuntu:~$ echo "Juan Navarro Maldonado | practica8 | Ubuntu 18.04" > info.txt
vagrant@ubuntu:~$ clear
vagrant@ubuntu:~$ cat info.txt
Juan Navarro Maldonado | practica8 | Ubuntu 18.04
vagrant@ubuntu:~$ sudo apt-get install fail2ban
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  libpython-stdlib libpython2.7-minimal libpython2.7-stdlib python
  python-minimal python2.7 python2.7-minimal python3-pyinotify whois
Suggested packages:
  mailx monit sqlite3 python-doc python-tk python2.7-doc binutils
  binfmt-support python-pyinotify-doc
The following NEW packages will be installed:
  fail2ban libpython-stdlib libpython2.7-minimal libpython2.7-stdlib python
  python-minimal python2.7 python2.7-minimal python3-pyinotify whois
0 upgraded, 10 newly installed, 0 to remove and 7 not upgraded.
Need to get 4370 kB of archives.
After this operation, 18.9 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu bionic-updates/main amd64 libpython2.7-minimal amd64 2.7.17-1~1
8.04ubuntu1.11 [335 kB]
Get:2 http://archive.ubuntu.com/ubuntu bionic-updates/main amd64 python2.7-minimal amd64 2.7.17-1~18.0
4ubuntu1.11 [1294 kB]
Get:3 http://archive.ubuntu.com/ubuntu bionic/main amd64 python-minimal amd64 2.7.15~rc1-1 [28.1 kB]
Get:4 http://archive.ubuntu.com/ubuntu bionic-updates/main amd64 libpython2.7-stdlib amd64 2.7.17-1~18
.04ubuntu1.11 [1919 kB]
Get:5 http://archive.ubuntu.com/ubuntu bionic-updates/main amd64 python2.7 amd64 2.7.17-1~18.04ubuntu1
.11 [248 kB]
Get:6 http://archive.ubuntu.com/ubuntu bionic/main amd64 libpython-stdlib amd64 2.7.15~rc1-1 [7620 B]
Get:7 http://archive.ubuntu.com/ubuntu bionic/main amd64 python amd64 2.7.15~rc1-1 [140 kB]
Get:8 http://archive.ubuntu.com/ubuntu bionic/universe amd64 fail2ban all 0.10.2-2 [329 kB]
Get:9 http://archive.ubuntu.com/ubuntu bionic/main amd64 python3-pyinotify all 0.9.6-1 [24.7 kB]
Get:10 http://archive.ubuntu.com/ubuntu bionic/main amd64 whois amd64 5.3.0 [43.7 kB]
Fetched 4370 kB in 1s (8081 kB/s)
Selecting previously unselected package libpython2.7-minimal:amd64.
(Reading database ... 60336 files and directories currently installed.)
Preparing to unpack .../0-libpython2.7-minimal_2.7.17-1~18.04ubuntu1.11_amd64.deb ...
Unpacking libpython2.7-minimal:amd64 (2.7.17-1~18.04ubuntu1.11) ...
Selecting previously unselected package python2.7-minimal.
Preparing to unpack .../1-python2.7-minimal_2.7.17-1~18.04ubuntu1.11_amd64.deb ...
Unpacking python2.7-minimal (2.7.17-1~18.04ubuntu1.11) ...
Selecting previously unselected package python-minimal.
Preparing to unpack .../2-python-minimal_2.7.15~rc1-1_amd64.deb ...
Unpacking python-minimal (2.7.15~rc1-1) ...
Selecting previously unselected package libpython2.7-stdlib:amd64.
Preparing to unpack .../3-libpython2.7-stdlib_2.7.17-1~18.04ubuntu1.11_amd64.deb ...
Unpacking libpython2.7-stdlib:amd64 (2.7.17-1~18.04ubuntu1.11) ...
```

Debemos de asegurarnos antes de que la configuración sea correcta, para ello nos metemos en el archivos de configuración que hemos copiado en /etc/fail2ban/jail.local y añadimos en el apartado de sshd enabled = true, también podemos ver en dicho fichero los parámetros por defecto, por ejemplo el bantime por defecto es de 10 minutos así como el máximo de intentos es 5, en mi caso voy a bajarlo a 2 intentos para realizar las pruebas posteriormente.



```
vagrant@ubuntu: ~ 
vagrant@ubuntu:~$ cat info.txt
Juan Navarro Maldonado | Practica 8 | Ubuntu 18.04
vagrant@ubuntu:~$ cat info.txt
Juan Navarro Maldonado | Practica 8 | Ubuntu 18.04
vagrant@ubuntu:~$ clear
vagrant@ubuntu:~$ cat info.txt
Juan Navarro Maldonado | Practica 8 | Ubuntu 18.04
vagrant@ubuntu:~$ cat /etc/fail2ban/jail.local | grep --color "enabled"
# enabled = true
# "enabled" enables the jails.
# true:  jail will be enabled and log files will get monitored for changes
# false: jail is not enabled
enabled = true
vagrant@ubuntu:~$ cat /etc/fail2ban/jail.local | grep --color "bantime"
# For example to change the default bantime for all jails and to enable the
# bantime = 1h
# "bantime" is the number of seconds that a host is banned.
bantime  = 10m
action_ = %(banaction)s[name=%(__name__)s, bantime="%(bantime)s", port="%(port)s", protocol="%(protoco
l)s", chain="%(chain)s"]
action_mw = %(banaction)s[name=%(__name__)s, bantime="%(bantime)s", port="%(port)s", protocol="%(proto
col)s", chain="%(chain)s"]
action_mwl = %(banaction)s[name=%(__name__)s, bantime="%(bantime)s", port="%(port)s", protocol="%(prot
ocol)s", chain="%(chain)s"]
action_xarf = %(banaction)s[name=%(__name__)s, bantime="%(bantime)s", port="%(port)s", protocol="%(pro
tocol)s", chain="%(chain)s"]
bantime  = 48h
bantime  = 1w
# consider low maxretry and a long bantime
bantime      = 1h
vagrant@ubuntu:~$
```

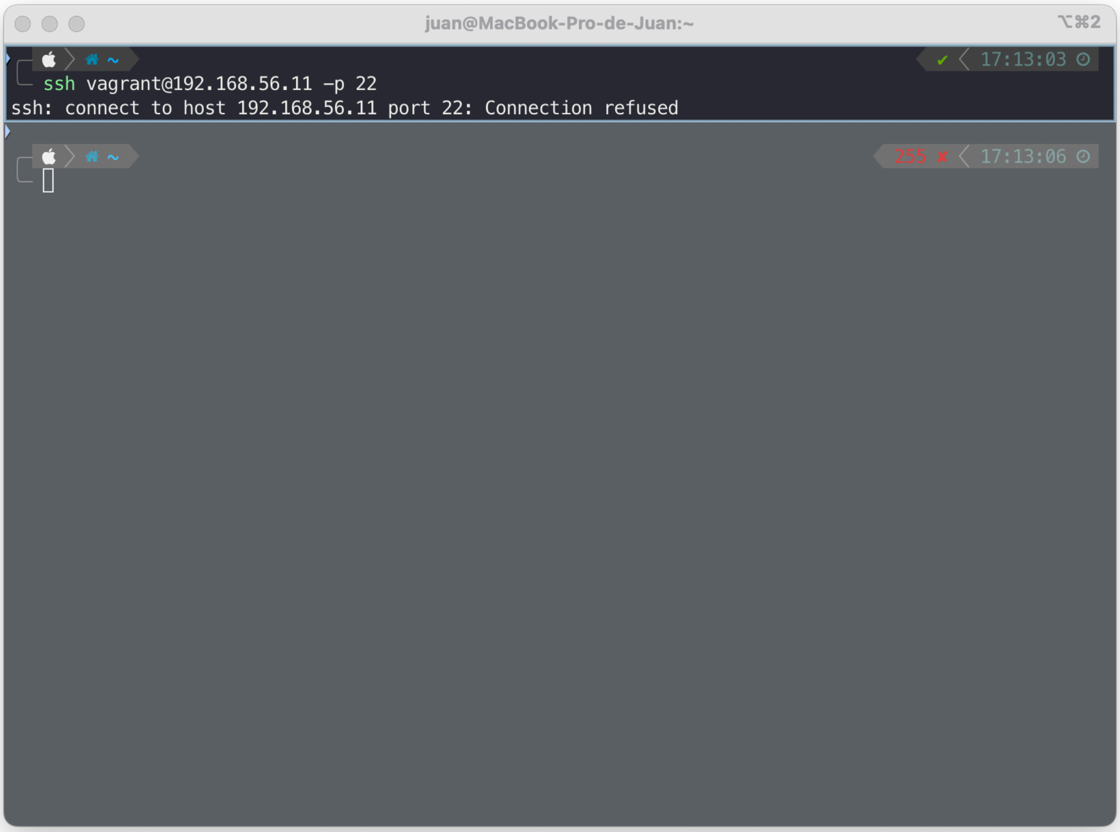
Ahora comprobamos el estado del cliente:



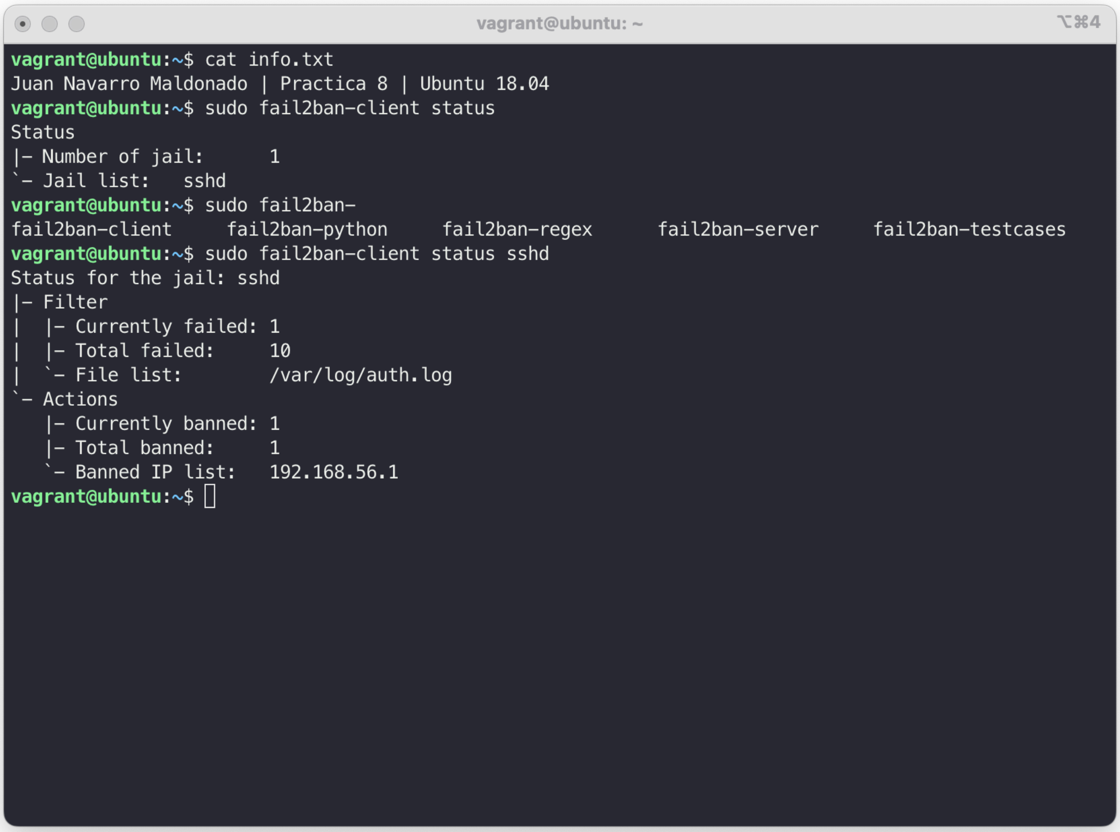
Por defecto podemos ver como se activa la jaula de ssh

Comprobación del acceso SSH

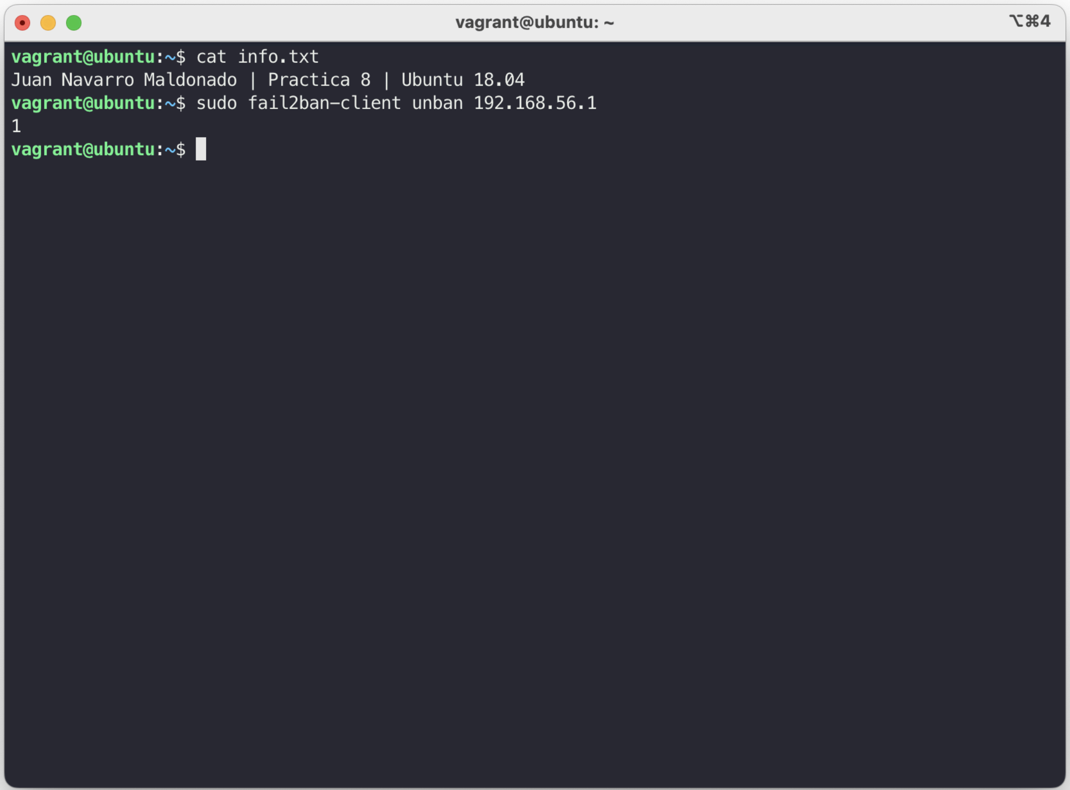
Para ello vamos a simular intentos fallidos desde mi máquina anfitriona, como vemos una vez que realizamos los intentos fallidos nos va a dar el siguiente problema:



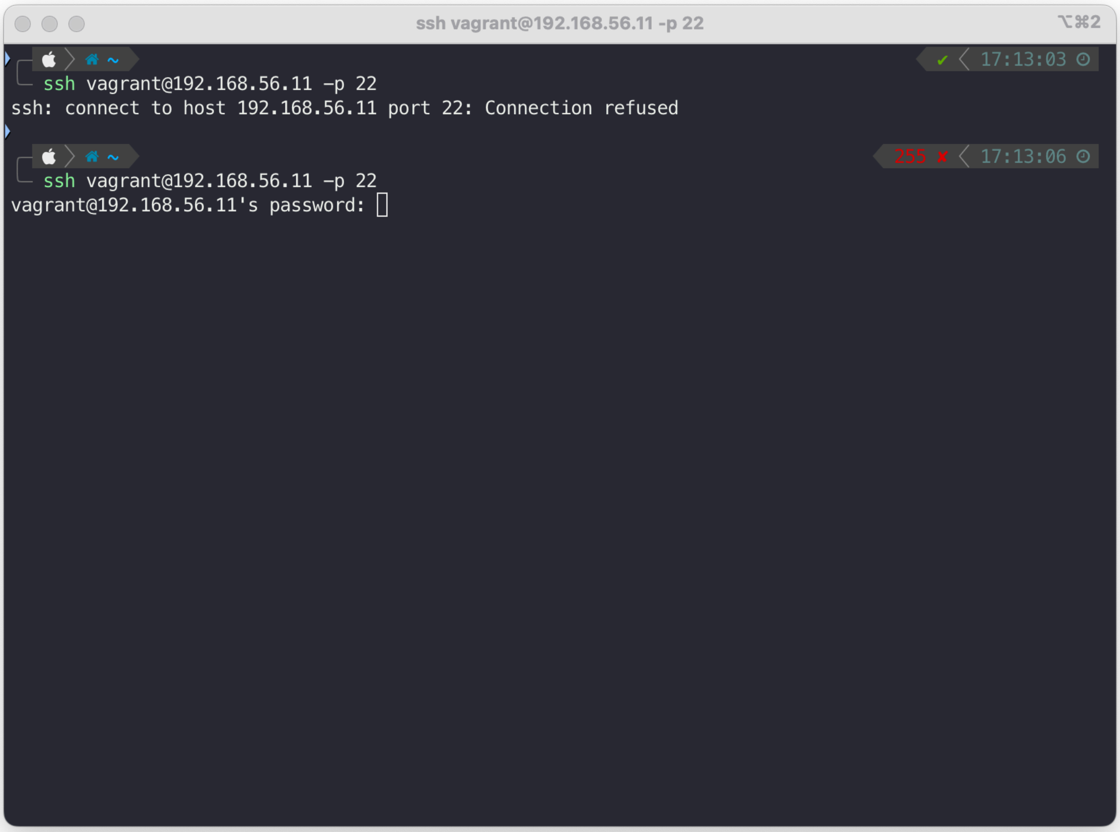
Y podemos ver en el status fail2ban para verificarlo:



Si queremos desbanear una dirección, con el siguiente comando:



Y podemos comprobar que se ha realizado correctamente probando a establecer una conexión por ssh:



2.- Instalación de Google Authenticator

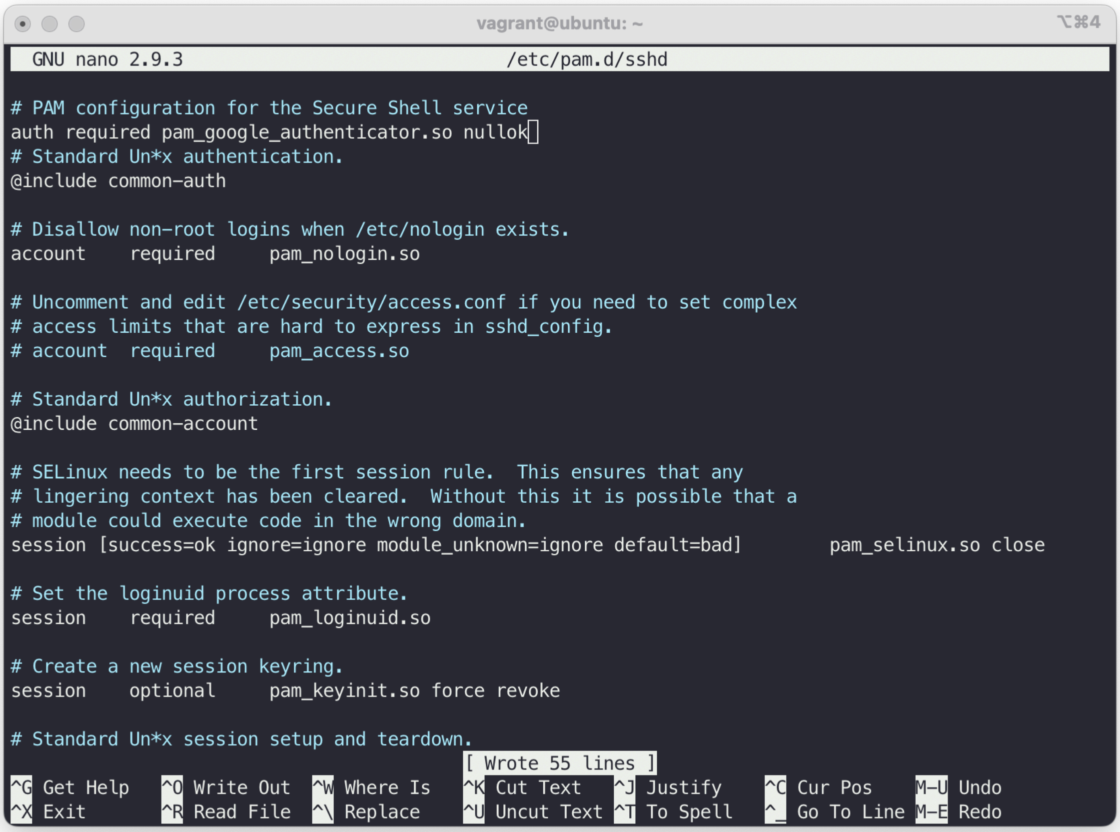
Instalación

Lo primero es instalar el módulo PAM siguiendo las indicaciones del guión con el siguiente comando:

```
vagrant@ubuntu: ~                                                          ⌥⌘4
vagrant@ubuntu:~$ cat info.txt
Juan Navarro Maldonado | Practica 8 | Ubuntu 18.04
vagrant@ubuntu:~$ sudo apt-get install libpam-google-authenticator
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  libqrencode3
The following NEW packages will be installed:
  libpam-google-authenticator libqrencode3
0 upgraded, 2 newly installed, 0 to remove and 7 not upgraded.
Need to get 56.8 kB of archives.
After this operation, 183 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu bionic/universe amd64 libqrencode3 amd64 3.4.4-1build1 [23.9 kB
]
Get:2 http://archive.ubuntu.com/ubuntu bionic/universe amd64 libpam-google-authenticator amd64 2017070
2-1 [32.9 kB]
Fetched 56.8 kB in 1s (84.7 kB/s)
Selecting previously unselected package libqrencode3:amd64.
(Reading database ... 61578 files and directories currently installed.)
Preparing to unpack .../libqrencode3_3.4.4-1build1_amd64.deb ...
Unpacking libqrencode3:amd64 (3.4.4-1build1) ...
Selecting previously unselected package libpam-google-authenticator.
Preparing to unpack .../libpam-google-authenticator_20170702-1_amd64.deb ...
Unpacking libpam-google-authenticator (20170702-1) ...
Setting up libqrencode3:amd64 (3.4.4-1build1) ...
Setting up libpam-google-authenticator (20170702-1) ...
Processing triggers for man-db (2.8.3-2ubuntu0.1) ...
Processing triggers for libc-bin (2.27-3ubuntu1.6) ...
vagrant@ubuntu:~$
```

Tenemos que editar el fichero /etc/pam.d/sshd y añadimos al comienzo del fichero la siguiente línea:

auth required pam_google_authenticator.so nullok

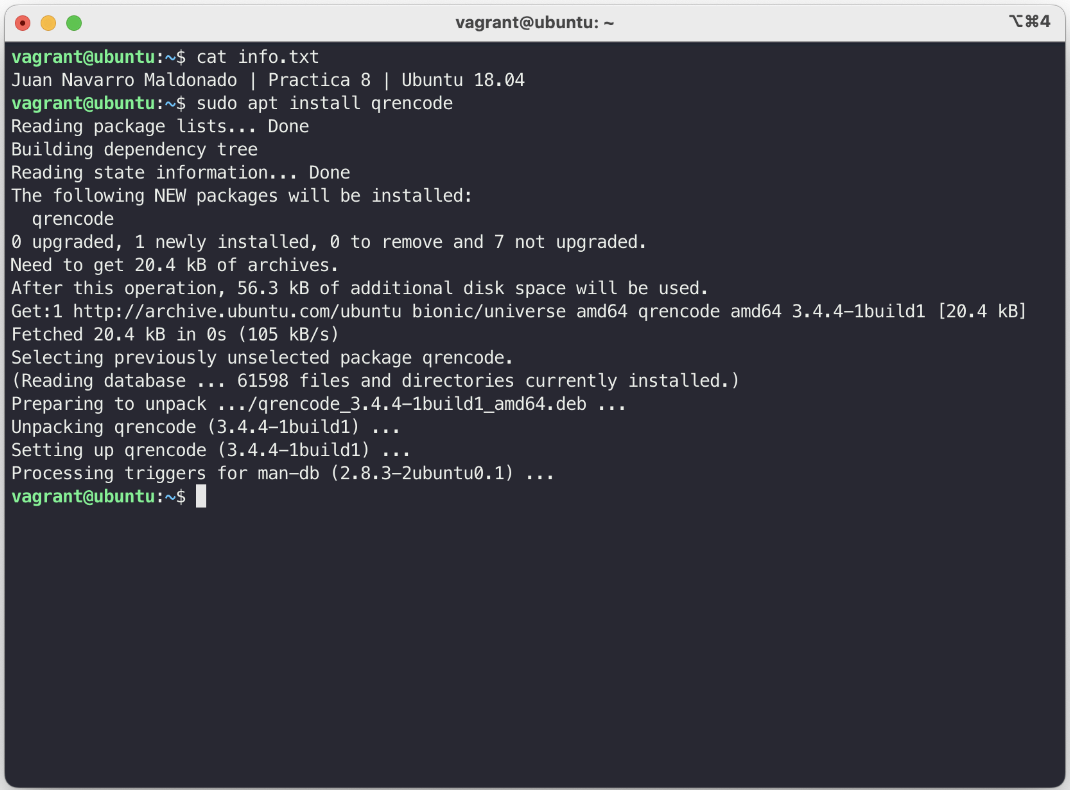


También debemos de modificar el fichero /etc/ssh/sshd_config para habilitar el challengeResponseAuthentication
Una vez cumplidos estos pasos debemos de reiniciar el servicio de ssh:

```
vagrant@ubuntu: ~
vagrant@ubuntu:~$ cat info.txt
Juan Navarro Maldonado | Practica 8 | Ubuntu 18.04
vagrant@ubuntu:~$ sudo nano /etc/pam.d/sshd
vagrant@ubuntu:~$ sudo nano /etc/ssh/sshd_config
vagrant@ubuntu:~$ sudo systemctl restart sshd
vagrant@ubuntu:~$ sudo systemctl restart ssh.service
vagrant@ubuntu:~$ sudo systemctl status sshd
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
   Active: active (running) since Thu 2024-11-28 16:23:59 UTC; 28s ago
  Process: 2506 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
 Main PID: 2512 (sshd)
    Tasks: 1 (limit: 2359)
   CGroup: /system.slice/ssh.service
           └─2512 /usr/sbin/sshd -D

Nov 28 16:23:59 ubuntu systemd[1]: Starting OpenBSD Secure Shell server...
Nov 28 16:23:59 ubuntu sshd[2512]: Server listening on 0.0.0.0 port 22.
Nov 28 16:23:59 ubuntu sshd[2512]: Server listening on :: port 22.
Nov 28 16:23:59 ubuntu systemd[1]: Started OpenBSD Secure Shell server.
vagrant@ubuntu:~$
```

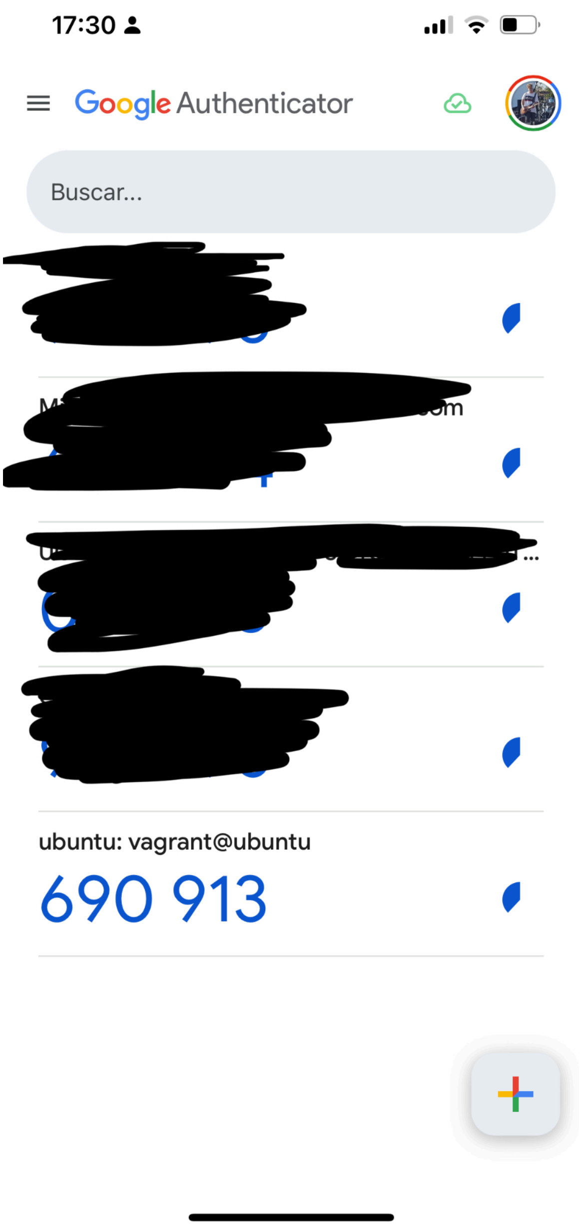
Y con esto pasaremos a la instalación de qrencode, que es una herramienta de línea de comandos que nos va a permitir generar códigos QR a partir de texto, esto junto con google Authenticator, nos va a permitir generar códigos QR para el sistema de verificación de dos factores, teniendo este QR la información del código temporal.

Para ello vamos a instalarlo con el siguiente comando:

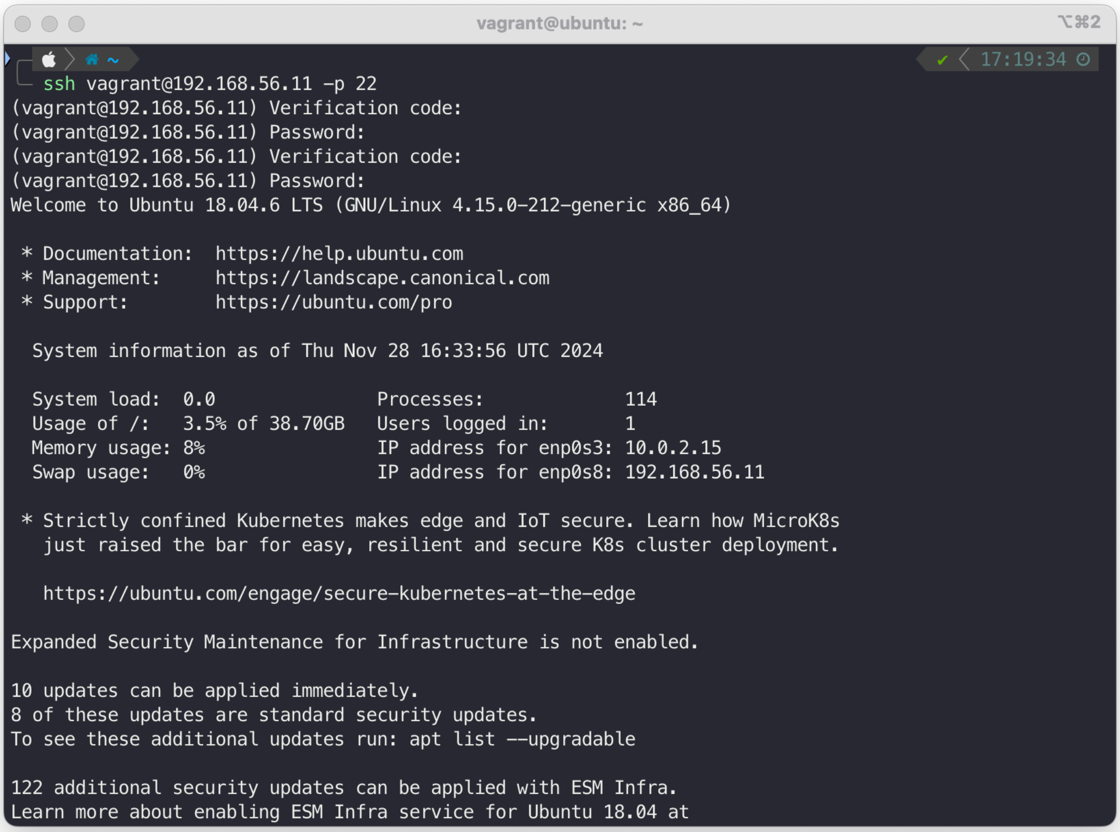


```
vagrant@ubuntu: ~
vagrant@ubuntu:~$ cat info.txt
Juan Navarro Maldonado | Practica 8 | Ubuntu 18.04
vagrant@ubuntu:~$ sudo apt install qrencode
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following NEW packages will be installed:
  qrencode
0 upgraded, 1 newly installed, 0 to remove and 7 not upgraded.
Need to get 20.4 kB of archives.
After this operation, 56.3 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu bionic/universe amd64 qrencode amd64 3.4.4-1build1 [20.4 kB]
Fetched 20.4 kB in 0s (105 kB/s)
Selecting previously unselected package qrencode.
(Reading database ... 61598 files and directories currently installed.)
Preparing to unpack .../qrencode_3.4.4-1build1_amd64.deb ...
Unpacking qrencode (3.4.4-1build1) ...
Setting up qrencode (3.4.4-1build1) ...
Processing triggers for man-db (2.8.3-2ubuntu0.1) ...
vagrant@ubuntu:~$
```

Finalmente ejecutaremos google-authenticator y nos dará nuestro código qr con la información, que escanear en el móvil para obtener mi código 2FA:


17:30
Google Authenticator
Buscar...
[redacted]
ubuntu: vagrant@ubuntu
690 913

Por último verificamos que todo se ha hecho correctamente entrando por ssh y vemos que nos pide el código de verificación junto con la contraseña para entrar por ssh



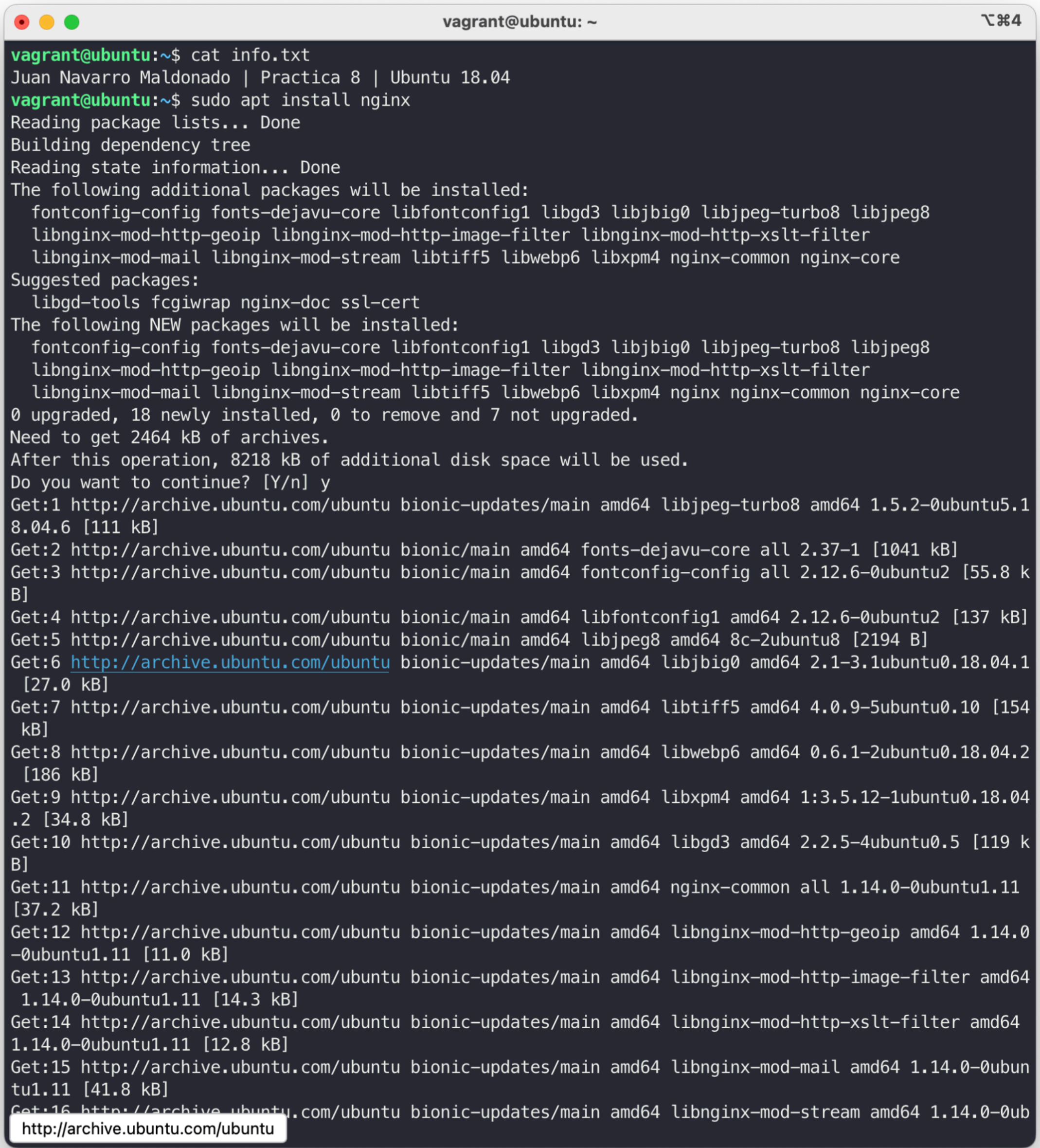
3.- (Opcional) Supervisión de NGINX mediante Fail2ban

Modificación del fichero jail.conf

Como ya he definido en la directiva global en los apartados anteriores en enabled, es decir que ya he activado todas las directivas, no tengo que realizar este paso.

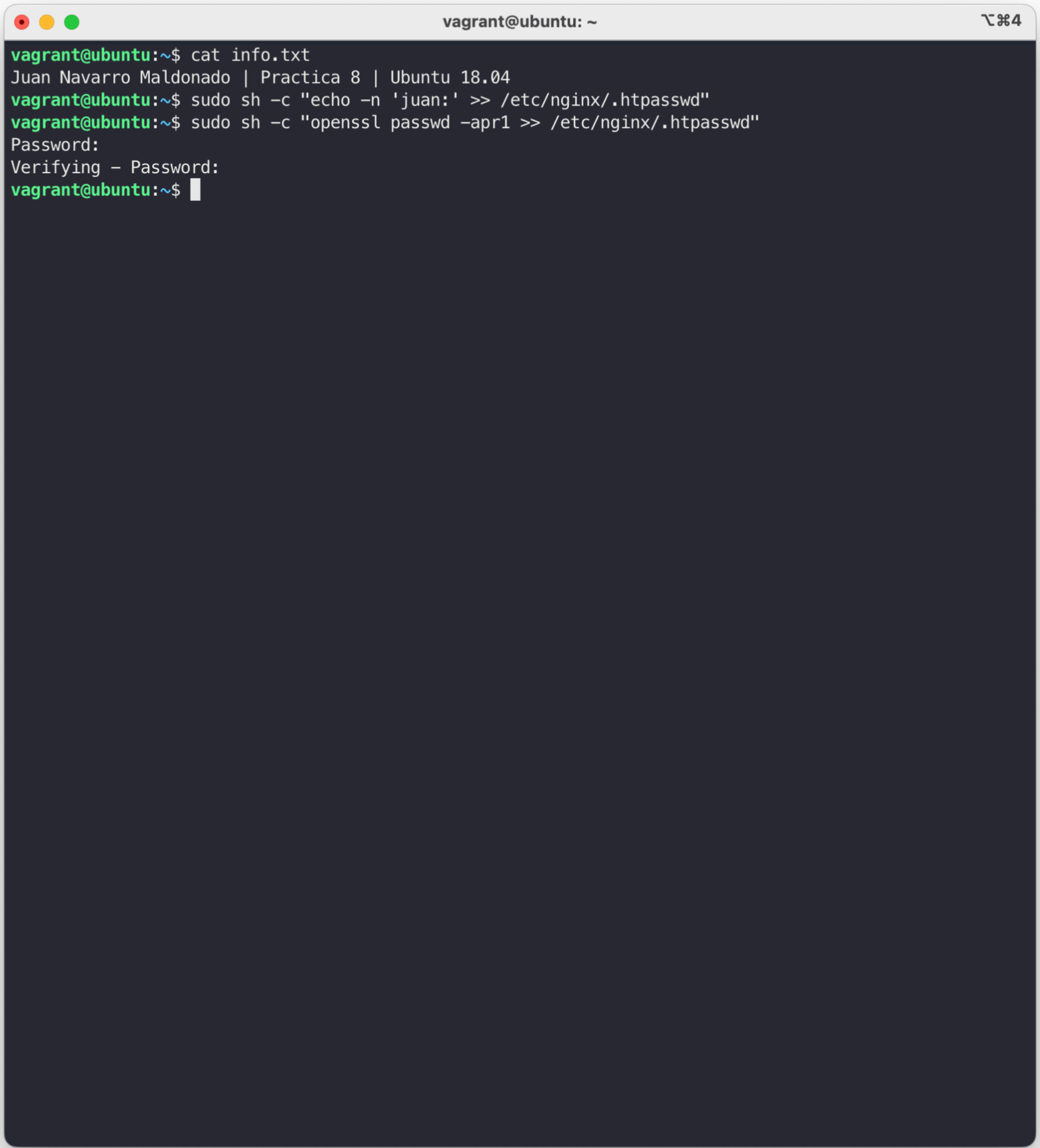
Instalación de NGINX en la máquina Ubuntu

Instalamos nginx con sudo apt install nginx en nuestra máquina virtual



Creación de una página con acceso mediante autenticación

Creamos el fichero /etc/nginx/.httpasswd con las claves:



También debemos de modificar el fichero /etc/nginx/sites-enabled/default:

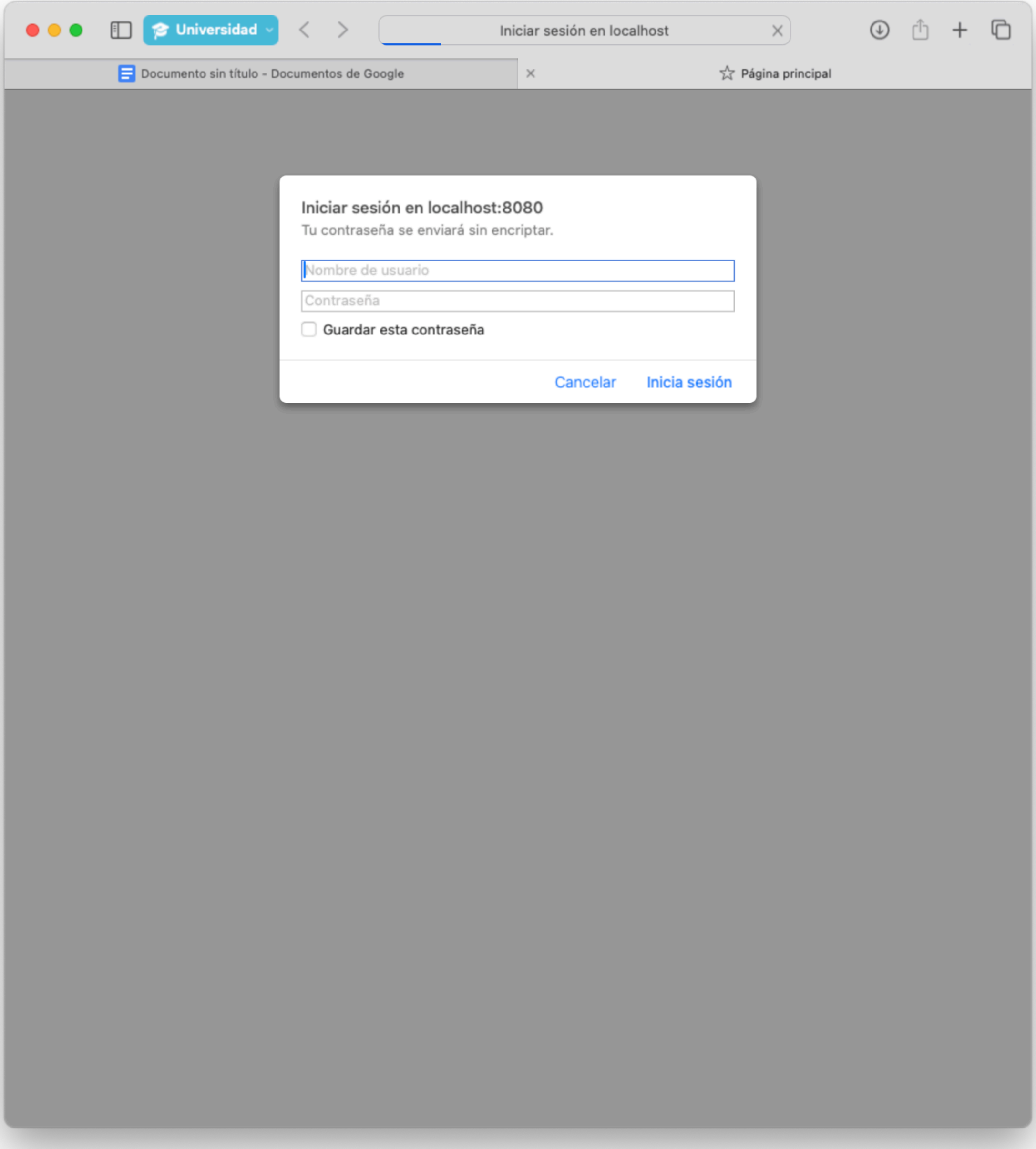
```
vagrant@ubuntu: ~                                                        ⌥⌘4
  GNU nano 2.9.3                  /etc/nginx/sites-enabled/default

#
server {
        listen 80 default_server;
        listen [::]:80 default_server;

        # SSL configuration
        #
        # listen 443 ssl default_server;
        # listen [::]:443 ssl default_server;
        #
        # Note: You should disable gzip for SSL traffic.
        # See: https://bugs.debian.org/773332
        #
        # Read up on ssl_ciphers to ensure a secure configuration.
        # See: https://bugs.debian.org/765782
        #
        # Self signed certs generated by the ssl-cert package
        # Don't use them in a production server!
        #
        # include snippets/snakeoil.conf;

        root /var/www/html;

        # Add index.php to the list if you are using PHP
        index index.html index.htm index.nginx-debian.html;

        server_name _;

        location / {
                # First attempt to serve request as file, then
                # as directory, then fall back to displaying a 404.
                try_files $uri $uri/ =404;
                auth_basic "Restricted Content";
                auth_basic_user_file /etc/nginx/.htpasswd;
        }

        # pass PHP scripts to FastCGI server
        #
        #location ~ \.php$ {
        #       include snippets/fastcgi-php.conf;
        #
        #       # With php-fpm (or other unix sockets):
        #       fastcgi_pass unix:/var/run/php/php7.0-fpm.sock;
        #       # With php-cgi (or other tcp sockets):
                                  [ Wrote 93 lines ]
^G Get Help   ^O Write Out  ^W Where Is   ^K Cut Text   ^J Justify    ^C Cur Pos    M-U Undo
^X Exit       ^R Read File  ^\ Replace    ^U Uncut Text ^T To Spell   ^_ Go To Line M-E Redo
```

y por ultimo reiniciamos el servicio

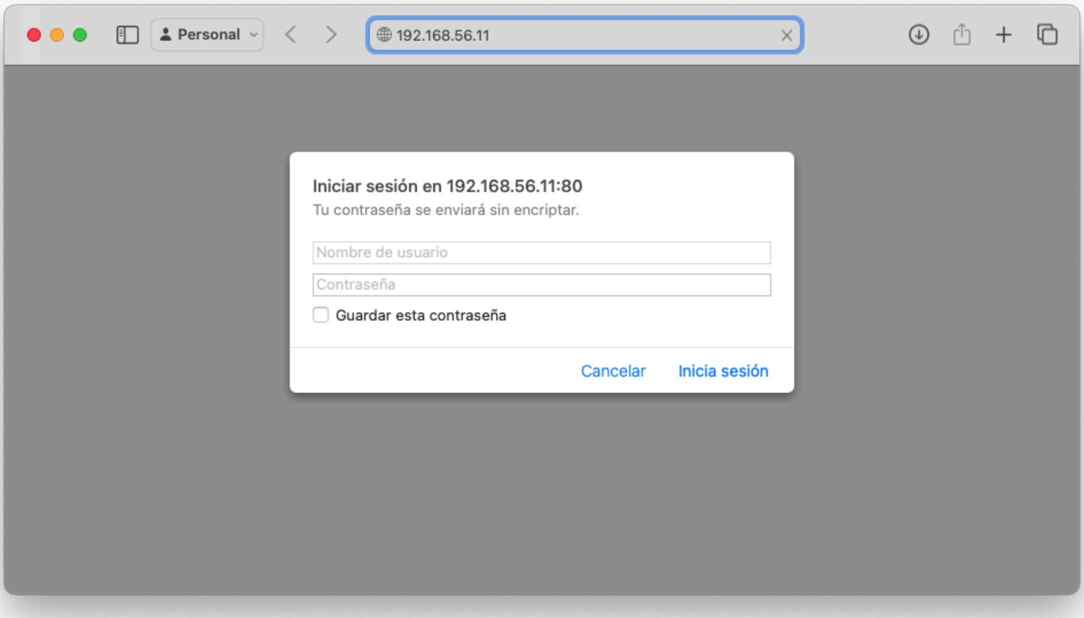
También pondremos un código básico para testear las credenciales con html

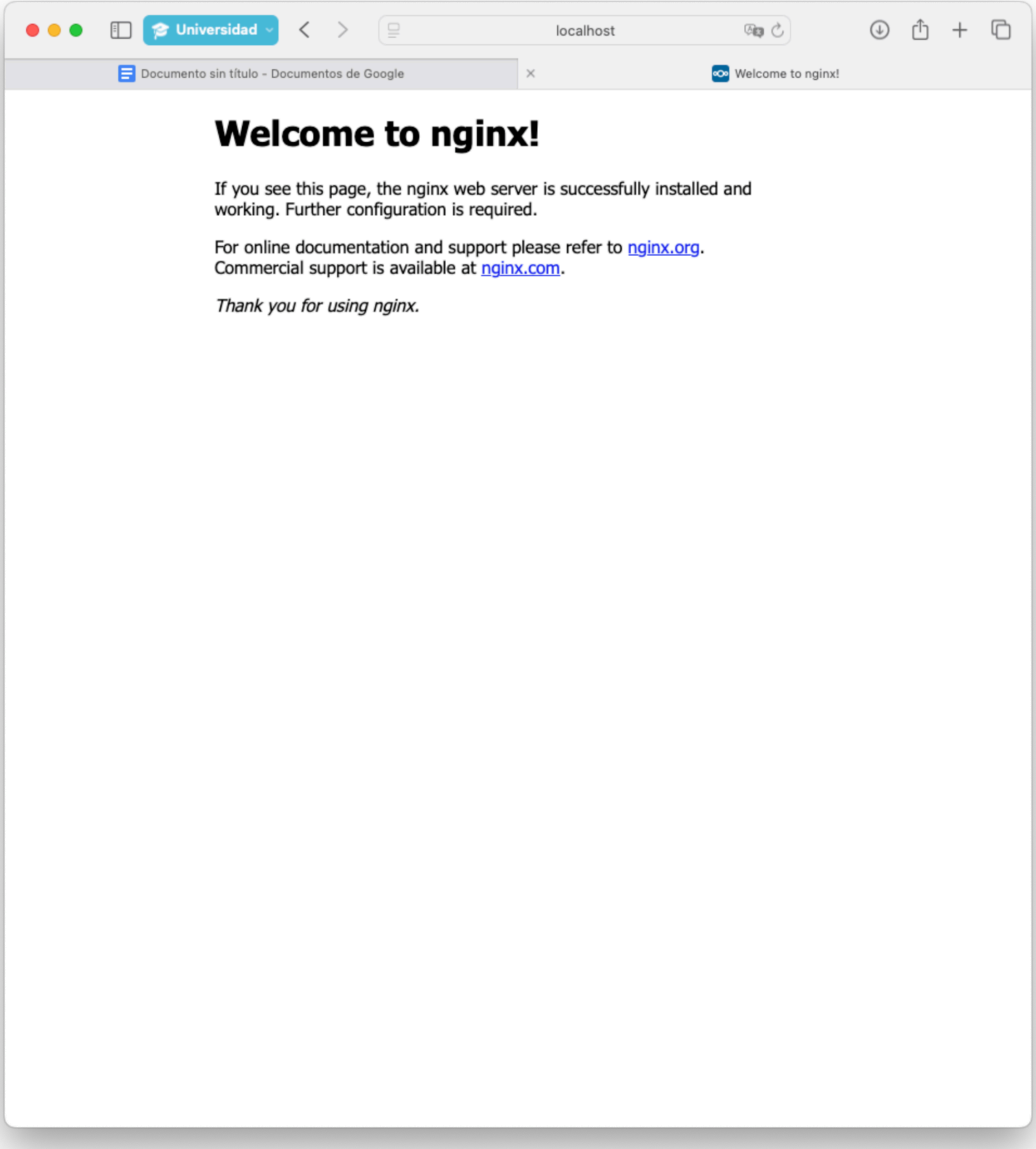
Comprobación entre nodos el bloqueo en los accesos a las páginas

Si entramos a la página podemos ver que nos pide los credenciales para acceder



Nota-> He entrado por localhost ya que en el vagrantfile tengo los puertos mapeados del 80 al 8080 aunque tambien se podria entrar con la dirección de nuestra máquina ubuntu como se muestra en la siguiente imagen

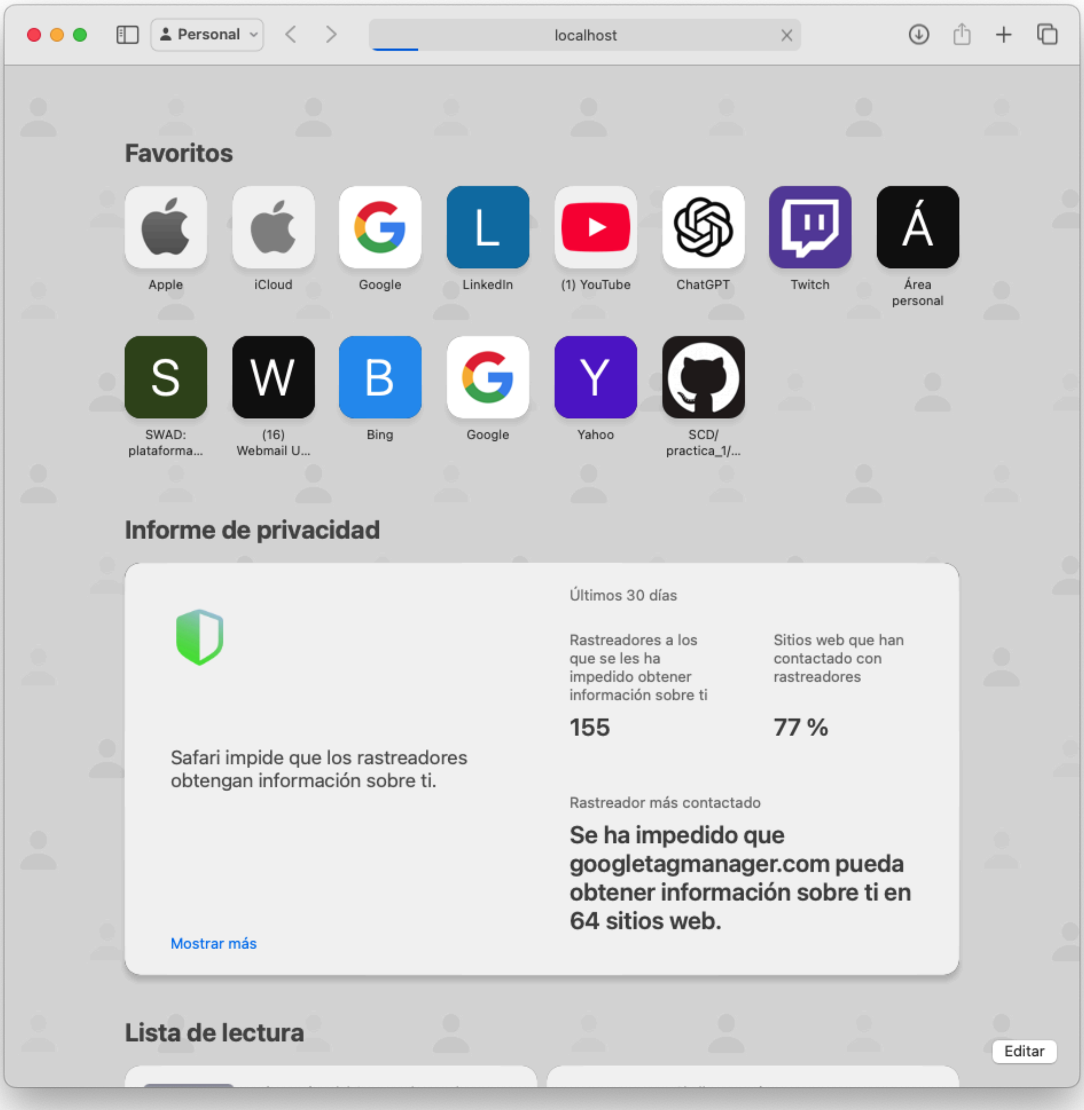



Universidad
localhost
Documento sin título - Documentos de Google
Welcome to nginx!
Welcome to nginx!
If you see this page, the nginx web server is successfully installed and working. Further configuration is required.
For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.
Thank you for using nginx.

También podemos verificar los fallos ingresando incorrectamente las credenciales

```
vagrant@ubuntu: ~                                                    ⌥⌘4
vagrant@ubuntu:~$ cat info.txt
Juan Navarro Maldonado | Practica 8 | Ubuntu 18.04
vagrant@ubuntu:~$ sudo fail2ban-client status nginx-http-auth
Status for the jail: nginx-http-auth
|- Filter
|  |- Currently failed: 0
|  |- Total failed:     5
|  `- File list:        /var/log/nginx/error.log
`- Actions
   |- Currently banned: 1
   |- Total banned:     1
   `- Banned IP list:   10.0.2.2
vagrant@ubuntu:~$
```

La página no cargará si intentamos acceder



4.- (Opcional) Utilizar la red TOR para acceder directamente a la máquina virtual

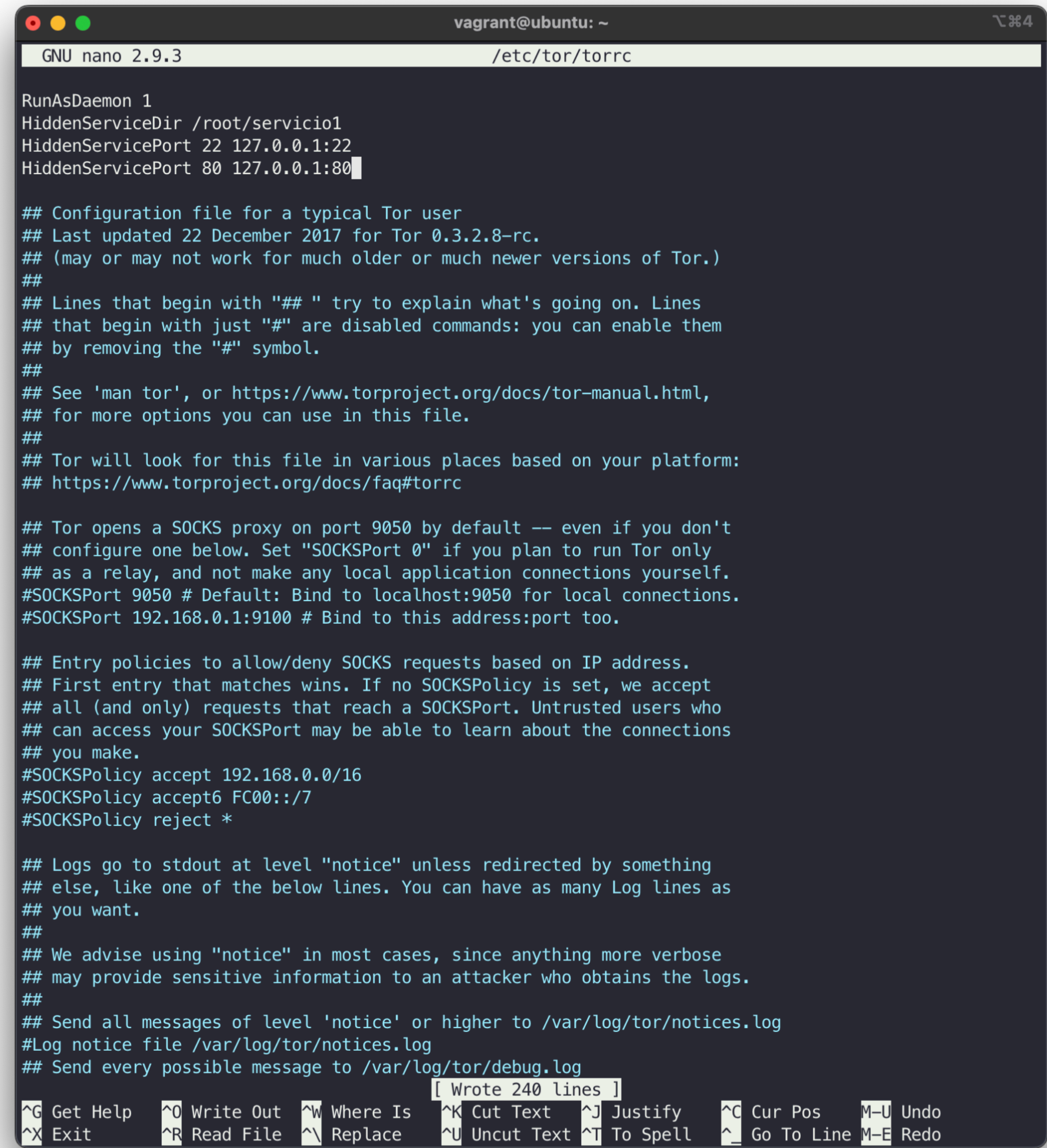
Instalación TOR

Instalamos tor con el comando sudo apt install tor

```
vagrant@ubuntu:~$ cat info.txt
Juan Navarro Maldonado | Practica 8 | Ubuntu 18.04
vagrant@ubuntu:~$ apt install tor
E: Could not open lock file /var/lib/dpkg/lock-frontend - open (13: Permission denied)
E: Unable to acquire the dpkg frontend lock (/var/lib/dpkg/lock-frontend), are you root?
vagrant@ubuntu:~$ sudo apt install tor
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  tor-geoipdb torsocks
Suggested packages:
  mixmaster torbrowser-launcher socat tor-arm apparmor-utils obfs4proxy
The following NEW packages will be installed:
  tor tor-geoipdb torsocks
0 upgraded, 3 newly installed, 0 to remove and 7 not upgraded.
Need to get 2111 kB of archives.
After this operation, 11.1 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu bionic/universe amd64 tor amd64 0.3.2.10-1 [1259 kB]
Get:2 http://archive.ubuntu.com/ubuntu bionic/universe amd64 torsocks amd64 2.2.0-2 [59.1 kB]
Get:3 http://archive.ubuntu.com/ubuntu bionic/universe amd64 tor-geoipdb all 0.3.2.10-1 [793 kB]
Fetched 2111 kB in 1s (1927 kB/s)
Selecting previously unselected package tor.
(Reading database ... 61842 files and directories currently installed.)
Preparing to unpack .../tor_0.3.2.10-1_amd64.deb ...
Unpacking tor (0.3.2.10-1) ...
Selecting previously unselected package torsocks.
Preparing to unpack .../torsocks_2.2.0-2_amd64.deb ...
Unpacking torsocks (2.2.0-2) ...
Selecting previously unselected package tor-geoipdb.
Preparing to unpack .../tor-geoipdb_0.3.2.10-1_all.deb ...
Unpacking tor-geoipdb (0.3.2.10-1) ...
Setting up tor (0.3.2.10-1) ...
Something or somebody made /var/lib/tor disappear.
Creating one for you again.
Something or somebody made /var/log/tor disappear.
Creating one for you again.
Created symlink /etc/systemd/system/multi-user.target.wants/tor.service → /lib/systemd/system/tor.serv
ice.
Setting up tor-geoipdb (0.3.2.10-1) ...
Setting up torsocks (2.2.0-2) ...
Processing triggers for systemd (237-3ubuntu10.57) ...
Processing triggers for man-db (2.8.3-2ubuntu0.1) ...
Processing triggers for ureadahead (0.100.0-21) ...
vagrant@ubuntu:~$
```

Configuracion del cliente

Para configurar el cliente debemos de editar el fisher /etc/tor/torrc y debemos de añadir las siguientes lineas al fichero:

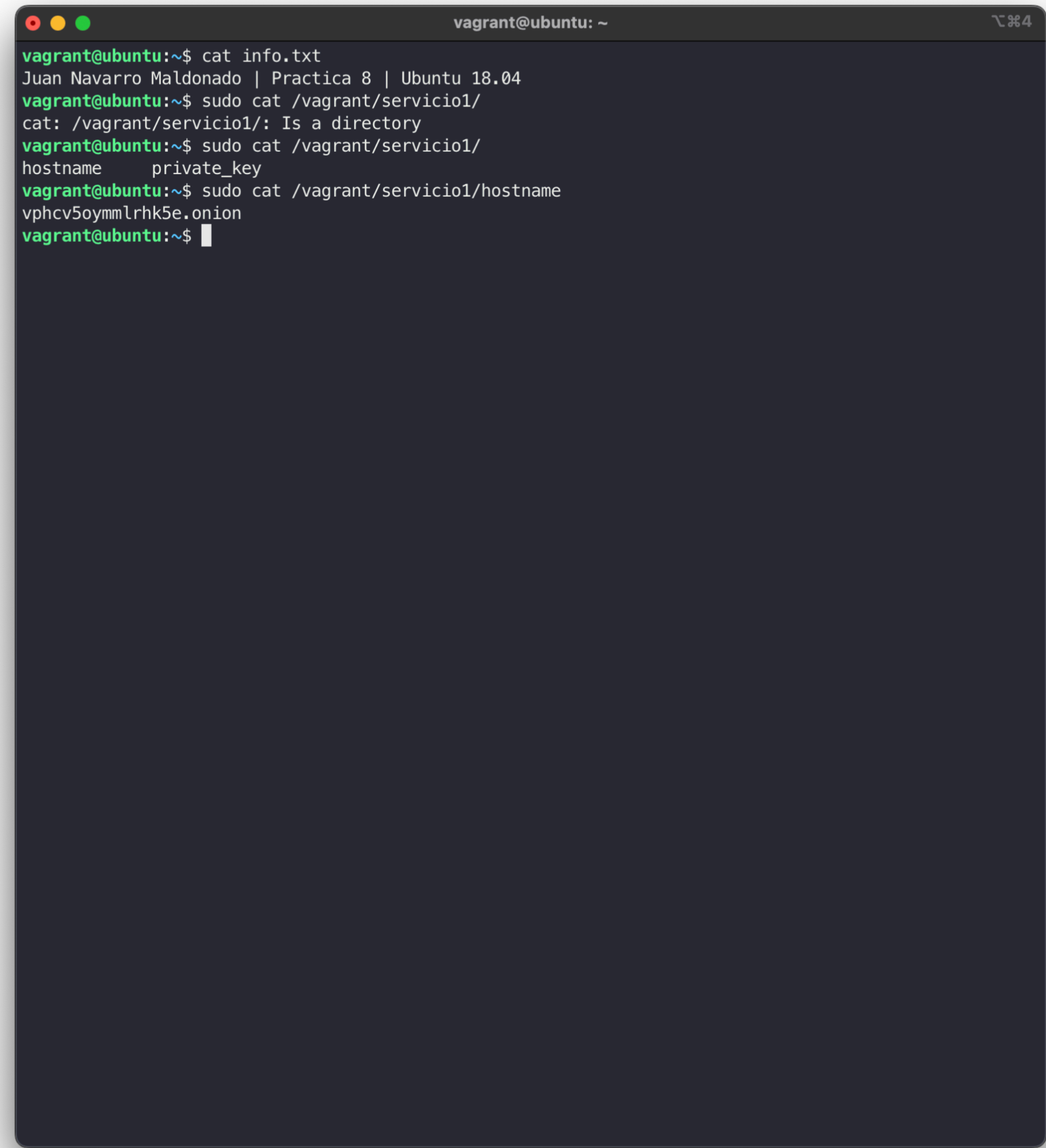


```
vagrant@ubuntu: ~                                                        ⌥⌘4
  GNU nano 2.9.3                        /etc/tor/torrc

RunAsDaemon 1
HiddenServiceDir /root/servicio1
HiddenServicePort 22 127.0.0.1:22
HiddenServicePort 80 127.0.0.1:80

## Configuration file for a typical Tor user
## Last updated 22 December 2017 for Tor 0.3.2.8-rc.
## (may or may not work for much older or much newer versions of Tor.)
##
## Lines that begin with "## " try to explain what's going on. Lines
## that begin with just "#" are disabled commands: you can enable them
## by removing the "#" symbol.
##
## See 'man tor', or https://www.torproject.org/docs/tor-manual.html,
## for more options you can use in this file.
##
## Tor will look for this file in various places based on your platform:
## https://www.torproject.org/docs/faq#torrc

## Tor opens a SOCKS proxy on port 9050 by default -- even if you don't
## configure one below. Set "SOCKSPort 0" if you plan to run Tor only
## as a relay, and not make any local application connections yourself.
#SOCKSPort 9050 # Default: Bind to localhost:9050 for local connections.
#SOCKSPort 192.168.0.1:9100 # Bind to this address:port too.

## Entry policies to allow/deny SOCKS requests based on IP address.
## First entry that matches wins. If no SOCKSPolicy is set, we accept
## all (and only) requests that reach a SOCKSPort. Untrusted users who
## can access your SOCKSPort may be able to learn about the connections
## you make.
#SOCKSPolicy accept 192.168.0.0/16
#SOCKSPolicy accept6 FC00::/7
#SOCKSPolicy reject *

## Logs go to stdout at level "notice" unless redirected by something
## else, like one of the below lines. You can have as many Log lines as
## you want.
##
## We advise using "notice" in most cases, since anything more verbose
## may provide sensitive information to an attacker who obtains the logs.
##
## Send all messages of level 'notice' or higher to /var/log/tor/notices.log
#Log notice file /var/log/tor/notices.log
## Send every possible message to /var/log/tor/debug.log
                              [ Wrote 240 lines ]
^G Get Help   ^O Write Out  ^W Where Is   ^K Cut Text   ^J Justify    ^C Cur Pos     M-U Undo
^X Exit       ^R Read File  ^\ Replace    ^U Uncut Text ^T To Spell   ^_ Go To Line  M-E Redo
```

Por último debemos de iniciar tor habiendo dado previamente los permisos necesario al directorio servicio1

```
vagrant@ubuntu: ~
vagrant@ubuntu:~$ cat info.txt
Juan Navarro Maldonado | Practica 8 | Ubuntu 18.04
vagrant@ubuntu:~$ sudo nano /etc/tor/torrc
vagrant@ubuntu:~$ sudo nano /etc/tor/torrc
vagrant@ubuntu:~$ tor
Nov 28 18:00:17.837 [notice] Tor 0.3.2.10 (git-0edaa32732ec8930) running on Linux with Libevent 2.1.8-
stable, OpenSSL 1.1.1, Zlib 1.2.11, Liblzma 5.2.2, and Libzstd 1.3.3.
Nov 28 18:00:17.837 [notice] Tor can't help you if you use it wrong! Learn how to be safe at https://w
ww.torproject.org/download/download#warning
Nov 28 18:00:17.837 [notice] Read configuration file "/etc/tor/torrc".
Nov 28 18:00:17.843 [warn] Directory /root/servicio1 cannot be read: Permission denied
Nov 28 18:00:17.843 [warn] Failed to parse/validate config: Failed to configure rendezvous options. Se
e logs for details.
Nov 28 18:00:17.843 [err] Reading config failed--see warnings above.
vagrant@ubuntu:~$ sudo tor
Nov 28 18:01:08.694 [notice] Tor 0.3.2.10 (git-0edaa32732ec8930) running on Linux with Libevent 2.1.8-
stable, OpenSSL 1.1.1, Zlib 1.2.11, Liblzma 5.2.2, and Libzstd 1.3.3.
Nov 28 18:01:08.694 [notice] Tor can't help you if you use it wrong! Learn how to be safe at https://w
ww.torproject.org/download/download#warning
Nov 28 18:01:08.694 [notice] Read configuration file "/etc/tor/torrc".
Nov 28 18:01:08.701 [notice] Scheduler type KIST has been enabled.
Nov 28 18:01:08.701 [notice] Opening Socks listener on 127.0.0.1:9050
Nov 28 18:01:08.701 [warn] Could not bind to 127.0.0.1:9050: Address already in use. Is Tor already ru
nning?
Nov 28 18:01:08.702 [warn] Failed to parse/validate config: Failed to bind one of the listener ports.
Nov 28 18:01:08.702 [err] Reading config failed--see warnings above.
vagrant@ubuntu:~$ sudo mkdir /root/servicio1
vagrant@ubuntu:~$ sudo chmod go-rwx /root/servicio1
vagrant@ubuntu:~$ ls -l /root/
ls: cannot open directory '/root/': Permission denied
vagrant@ubuntu:~$ sudo ls -l /root/
total 4
drwx------ 2 root root 4096 Nov 28 18:03 servicio1
vagrant@ubuntu:~$ sudo systemctl start tor
vagrant@ubuntu:~$ sudo systemctl enable tor
Synchronizing state of tor.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable tor
vagrant@ubuntu:~$ sudo systemctl status tor
● tor.service - Anonymizing overlay network for TCP (multi-instance-master)
   Loaded: loaded (/lib/systemd/system/tor.service; enabled; vendor preset: enabled)
   Active: active (exited) since Thu 2024-11-28 17:56:54 UTC; 7min ago
 Main PID: 3881 (code=exited, status=0/SUCCESS)
    Tasks: 0 (limit: 2359)
   CGroup: /system.slice/tor.service

Nov 28 17:56:54 ubuntu systemd[1]: Starting Anonymizing overlay network for TCP (multi-instance-master
Nov 28 17:56:54 ubuntu systemd[1]: Started Anonymizing overlay network for TCP (multi-instance-master)
lines 1-9/9 (END)
```

Podemos ver los servicios que nos ha instalador tor ahora



Con estos pasos ya estaría configurado tor en nuestra máquina virtual.